C4.1 How is data used to choose a material for a particular use?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Our society uses a large range of materials and products developed by chemists. Chemists assess materials by measuring their physical properties, and use data to compare different materials and to match materials to the specification of a useful product (IaS4). Composites have a very broad range of uses as they allow the properties of several materials to be combined. Composites may have materials combined on a bulk scale (for example using steel to reinforce concrete) or have nanoparticles incorporated in a material or embedded in a matrix. The range of uses of metals has been extended by the development of alloys. Alloys have different properties to pure metals due to the disruption of the metal lattice by atoms of different sizes. Chemists can match an alloy to the specification of properties for a new product.	1. compare quantitatively the physical properties of glass and clay ceramics, polymers, composites and metals, including melting point, softening temperature (for polymers), electrical conductivity, strength (in tension or compression), stiffness, flexibility, brittleness, hardness, density, ease of reshaping 2. explain how the properties of materials are related to their uses and select appropriate materials given details of the usage required 3. describe the composition of some important alloys in relation to their properties and uses, including steel *(separate science only)*	**Practical work**: • Practical investigation of a range of materials leading to classification into categories. **Ideas about Science:** • The range of materials developed by chemists enhances the quality of life. (IaS4) • Use and limitations of a model to represent alloy structure. (IaS3)